SFU

Downloading many-qubit entanglement from continuous-variable cluster state

Zhi Han, **Kero Lau** (Simon Fraser University, Physics)

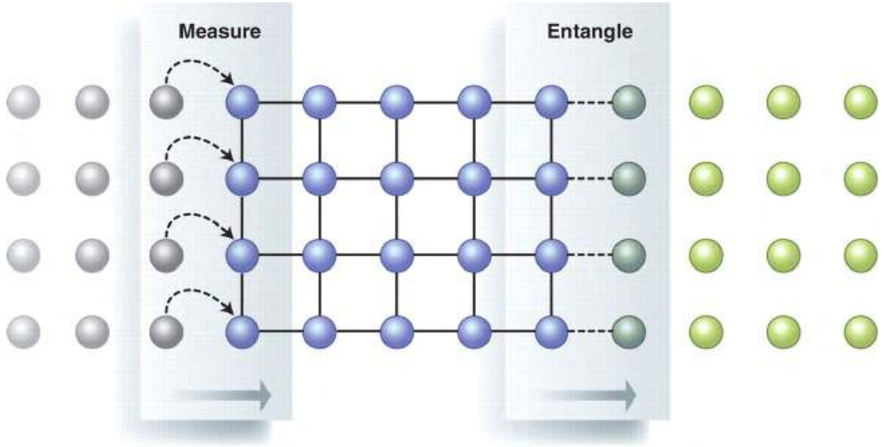
NSERC CRSNG

mitacs

CANADA RESEARCH CHAIRS
CHAIRES DE RECHERCHE DU CANADA

We all want many-body entanglement, right?

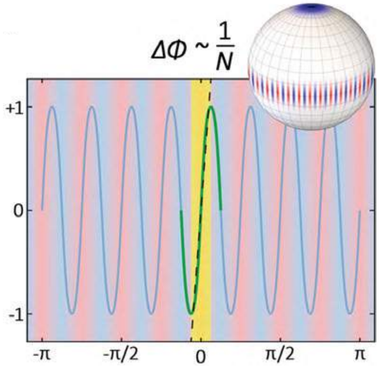
O'Brien, *Science* **318**, 1567 (2007)



Quantum computing

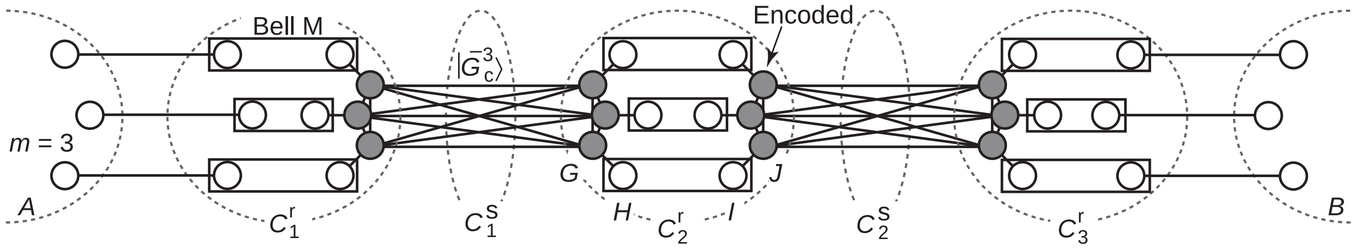
Ye & Zoller, *PRL* **132**, 190001 (2024)





Quantum sensing

Azuma *et al.*, *Nat. Comm.* **6**, 6787 (2007)

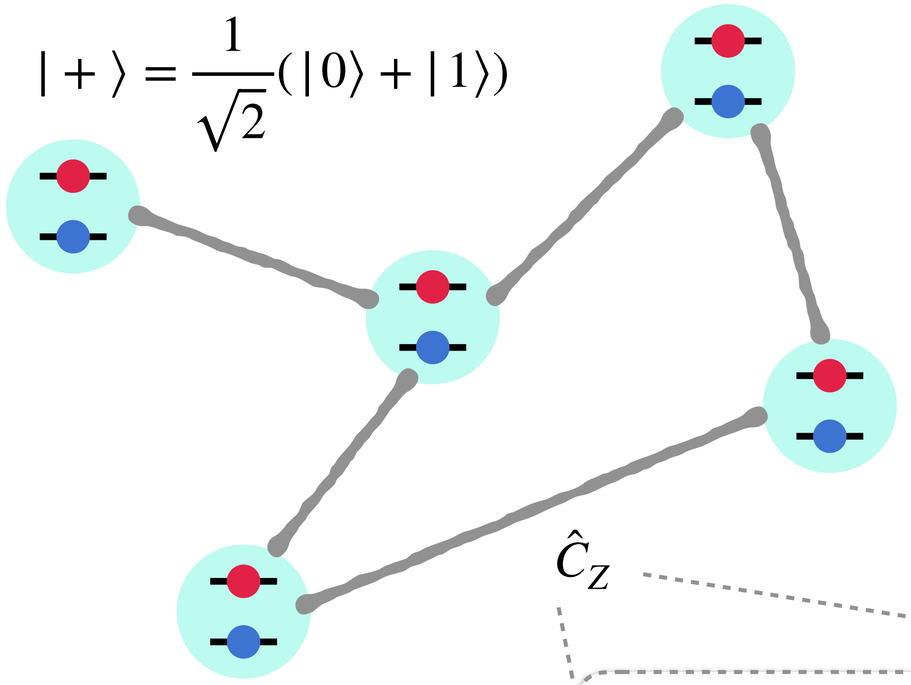


Quantum communication (repeater)

Cluster state: universal resources

Cluster/Graph state

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$



$\hat{C}_Z$

$$|00\rangle \to |00\rangle \quad |01\rangle \to |01\rangle$$
$$|10\rangle \to |10\rangle \quad |11\rangle \to -|11\rangle$$

Measurement

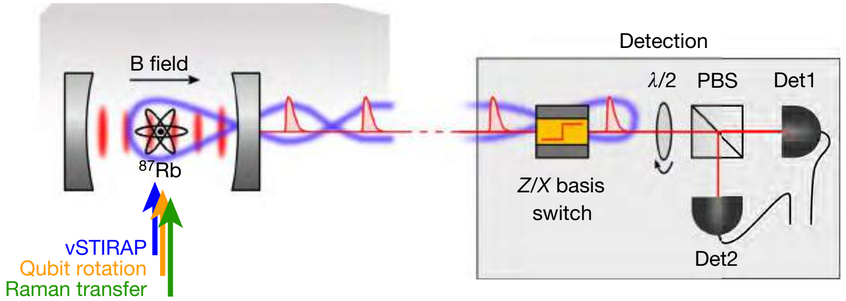
Any many-body entangled state

$$|\Psi\rangle$$

- Universal quantum computing resources
- Surface code
- Blind quantum computing

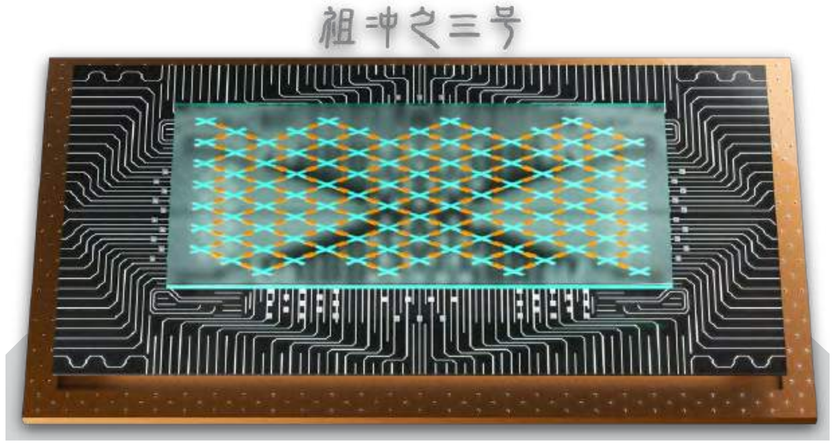
How much qubit cluster states do we have?

Thomas *et al.*, *Nature* **608**, 677 (2022)



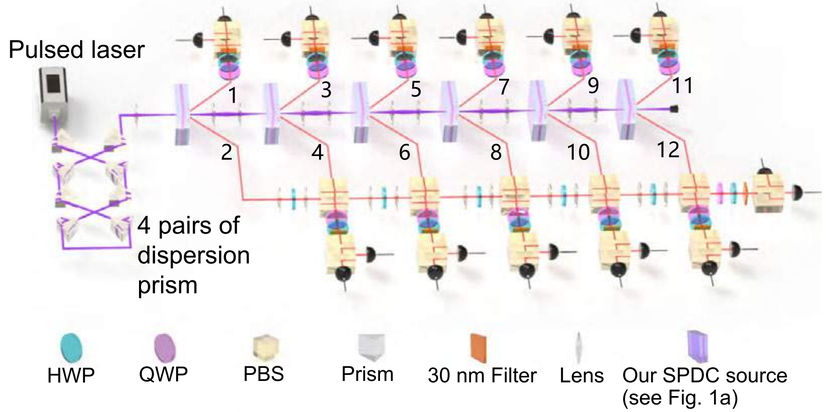
Photon (emitter): 14

Jiang *et al.*, *Nat. Phys.* **22**, 430 (2026)



Superconductor: 95

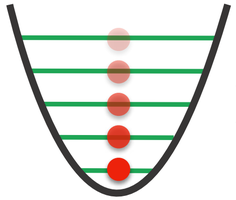
Zhong *et al.*, *PRL* **121**, 250505 (2018)



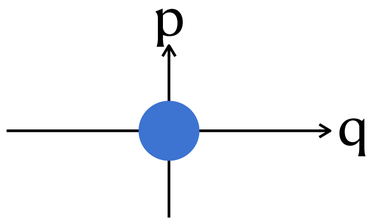
Photon (SPDC): 12

Can we get more entanglement if we go beyond qubit?

Continuous-variable (CV) cluster state

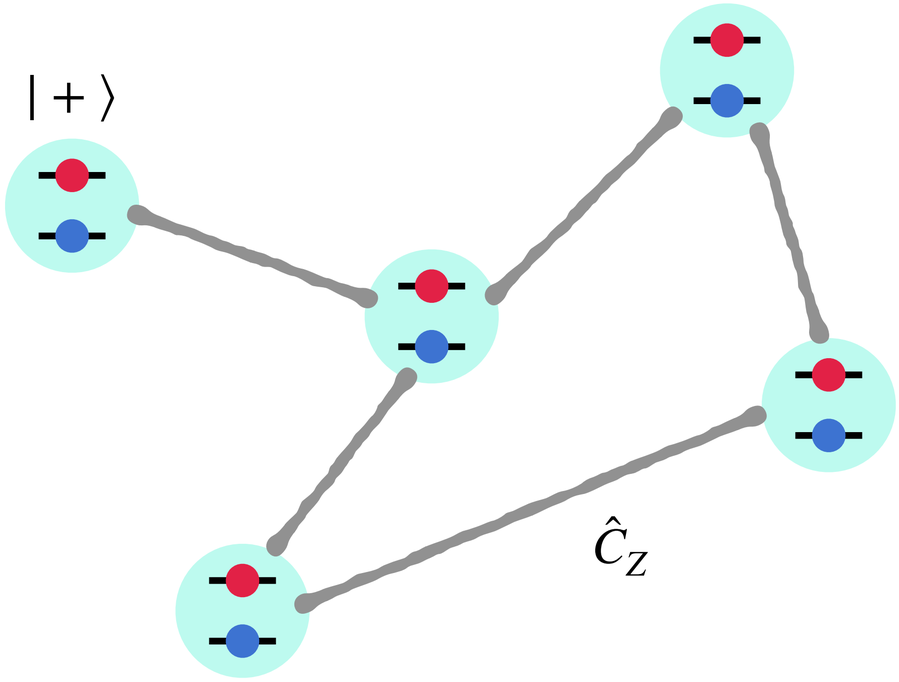


Bosonic mode = harmonic oscillator

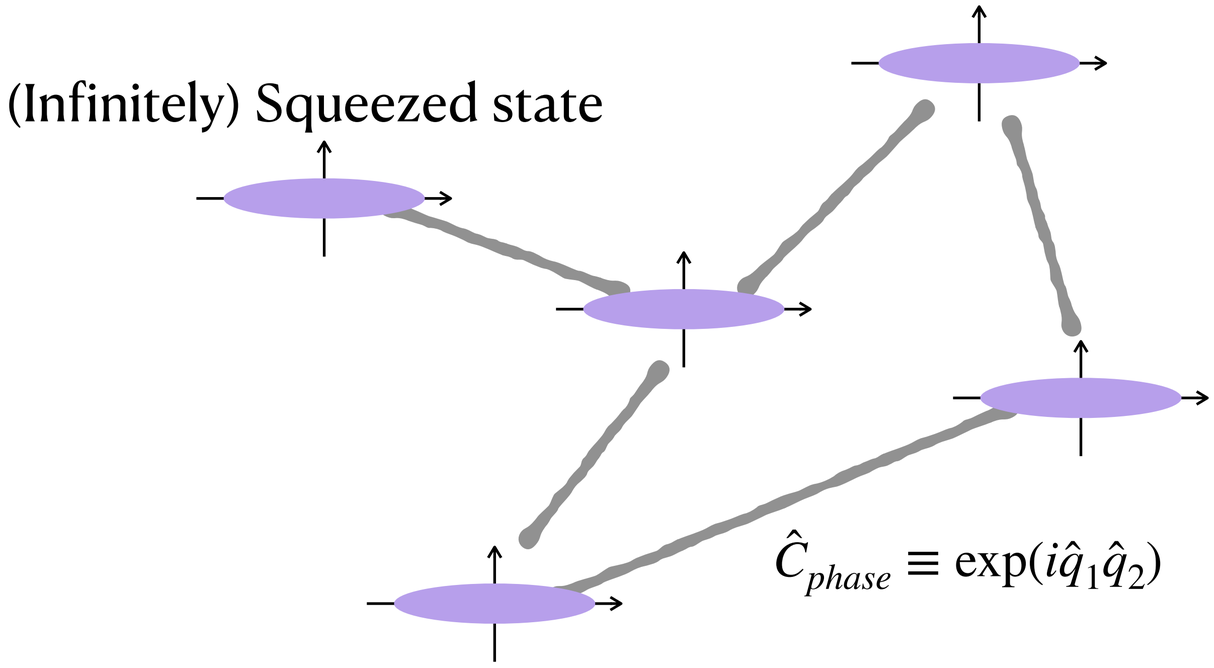


State as continuous function

Qubit cluster state



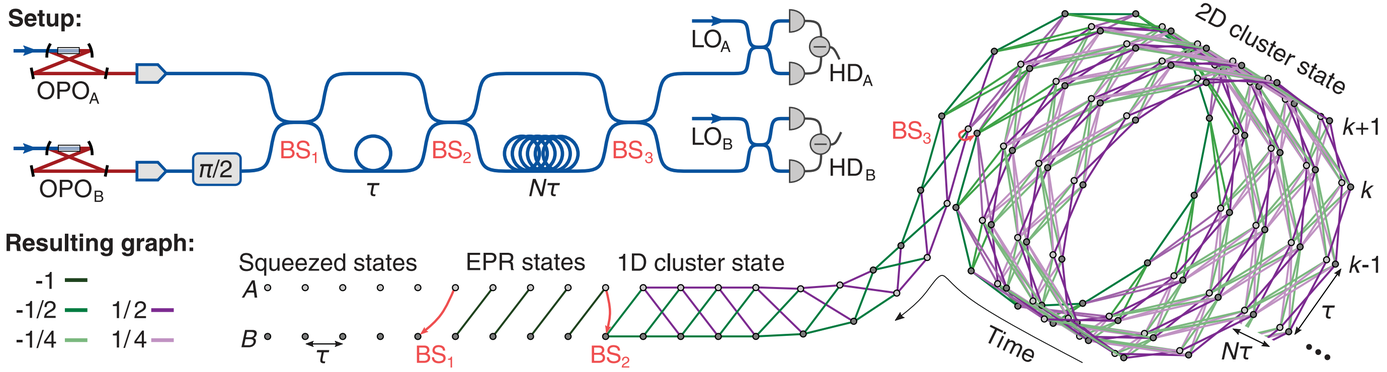
Continuous-Variable cluster state



How much CV entanglement do we have?

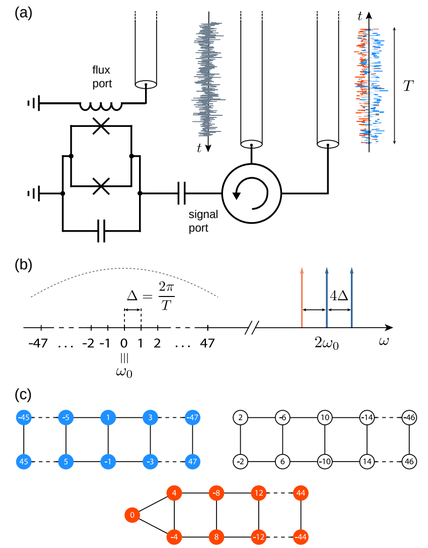
Yokoyama *et al.*, *Nat. Photonics* **7**, 982 (2013)
Yoshikawa *et al.*, *APL Photonics* **1**, 060801 (2016)

Larsen *et al.*, *Science* **366**, 369 (2019)
Asavanant *et al.*, *Science* **366**, 373 (2019)



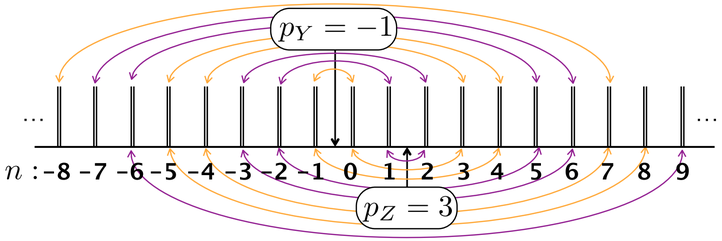
Optical pulses: 1 million (1D), 30000 (2D)

Lingua *et al.*, *PRL* **134**, 183602 (2025)



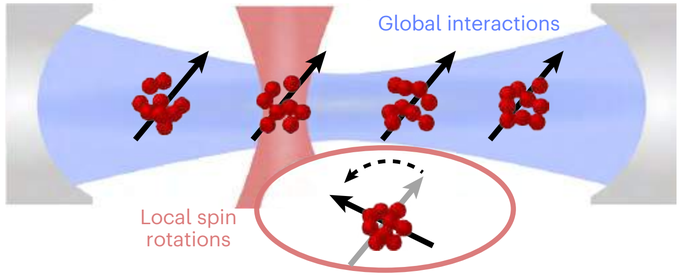
Microwave: 94

Chen *et al.*, *PRL* **112**, 120505 (2014)



Optical frequency mode: 60

Cooper *et al.*, *Nat. Phys.* **20**, 770 (2024)

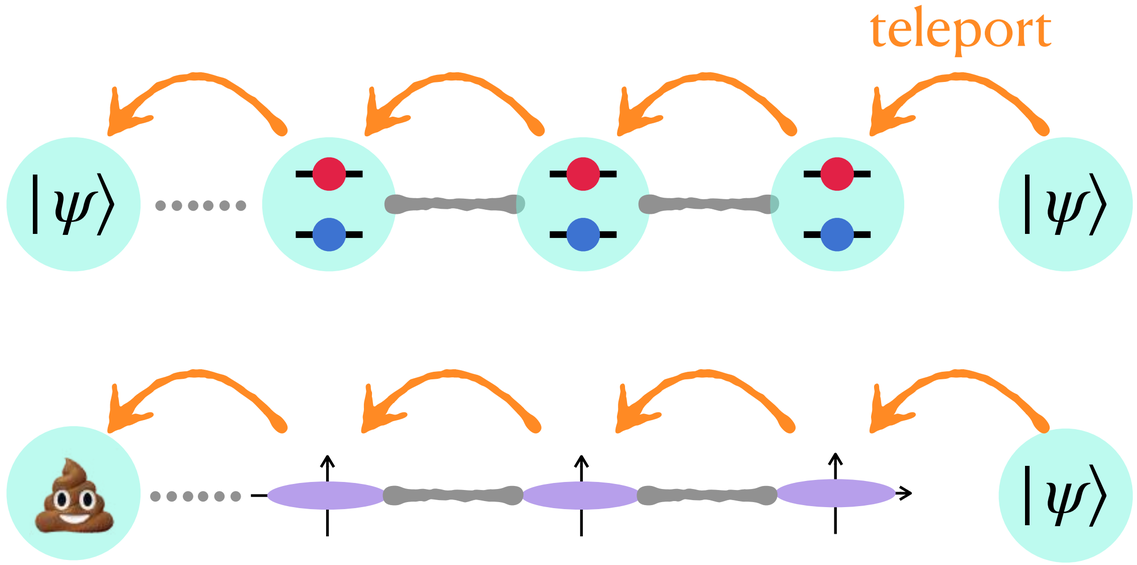


Atomic ensemble: 4

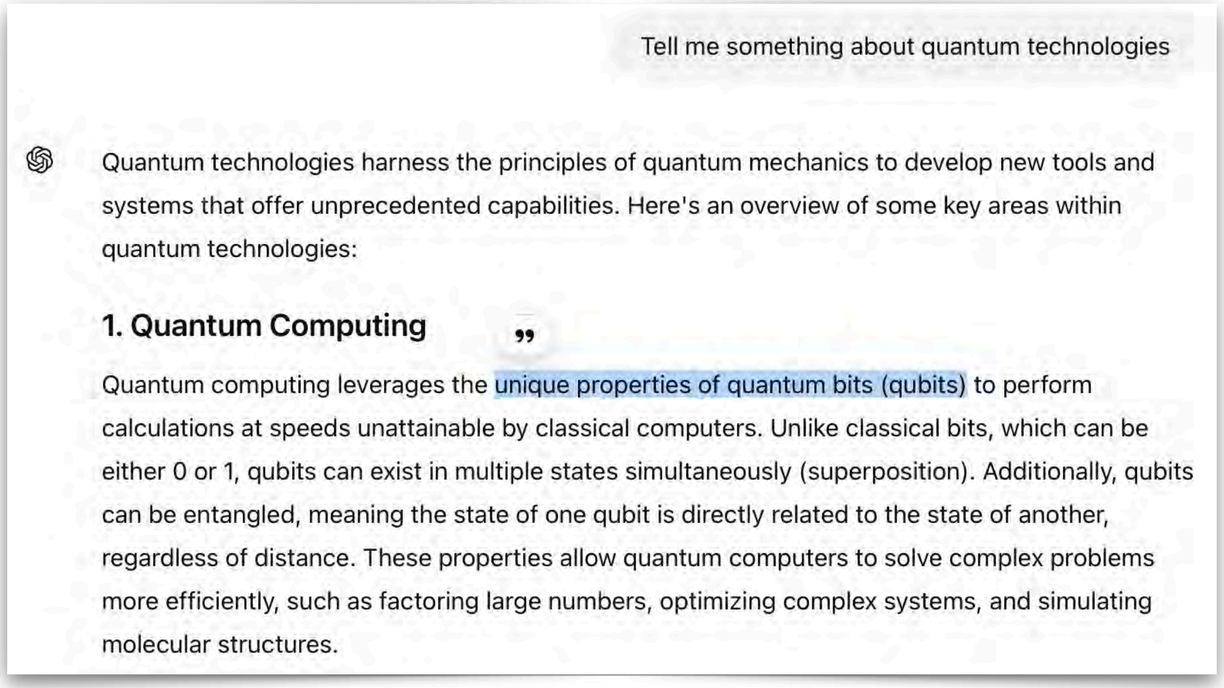
So we have many-body entanglement, we can live happily hereafter?

Fundamental & Practical problems of CV

1. Gaussian entanglement must introduce error (unless you have infinite energy)



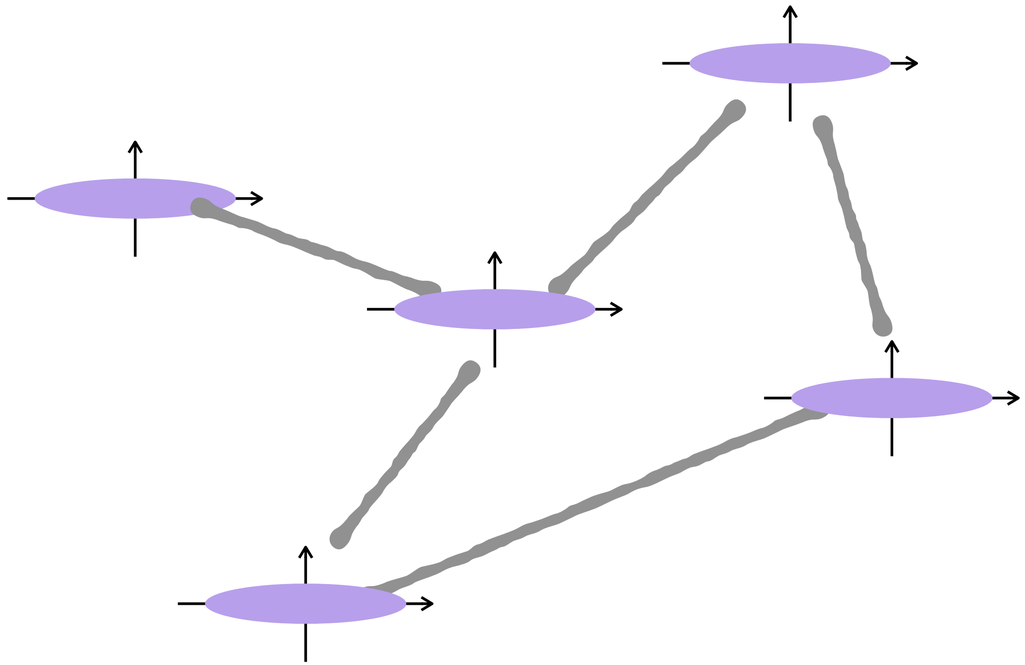
2. Most quantum technologies are based on qubits



Ohliger, Kieling, Eisert, *PRA* **82**, 042336 (2010)

Downloading many-qubit entanglement from continuous-variable cluster state

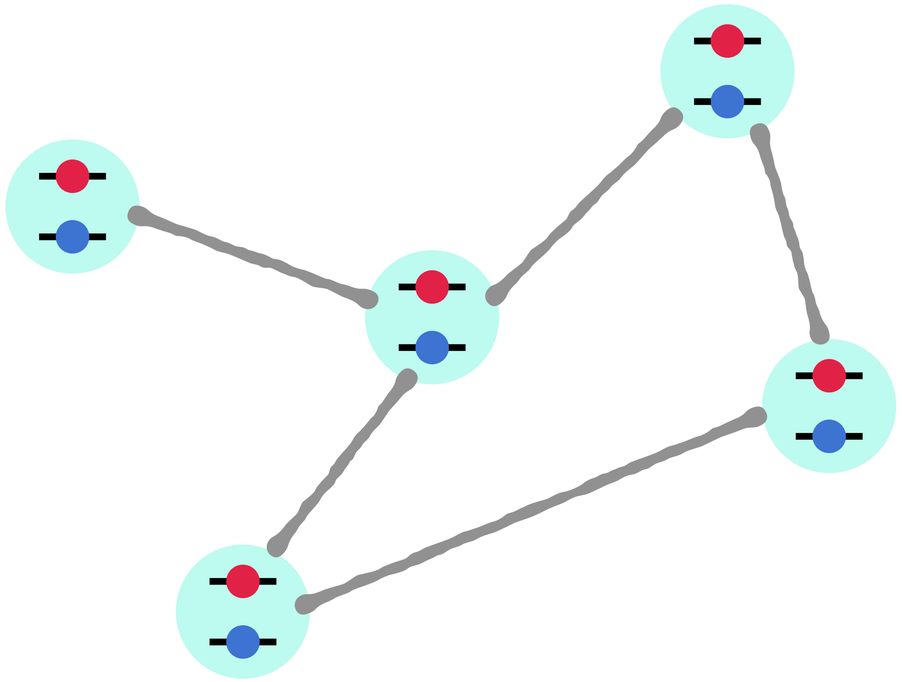
CV cluster state



Efficiently generated,

~~but not too useful~~

Qubit cluster state

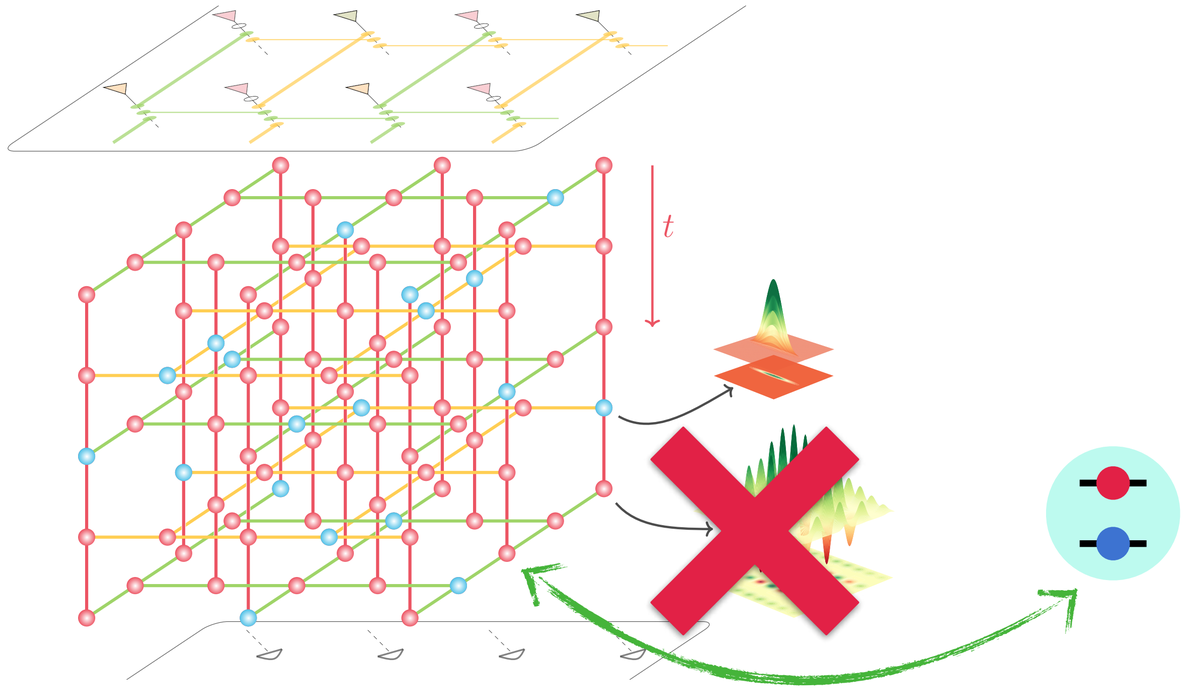


Lots of applications,

~~but hard too make~~

Zhi Han, Hoi-Kwan Lau, *Physical Review Letters* **136**, 040202 (2026)

Fixing Gaussian problems with GKP states?



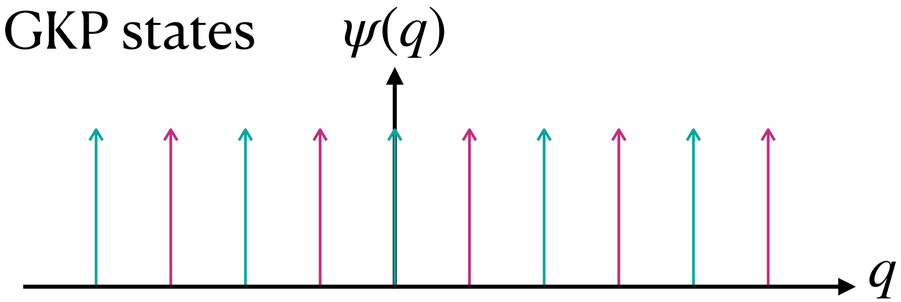
Menicucci, *PRL* **112**, 120504 (2014)
Pantaleoni *et al.*, *PRL* **125**, 040501 (2020)
Bourassa *et al.*, *Quantum* **5**, 392 (2021)

- Qubit
- Ideal
- Available in hybrid systems, e.g. trapped ions, circuit QED

Ideal (unphysical) GKP states

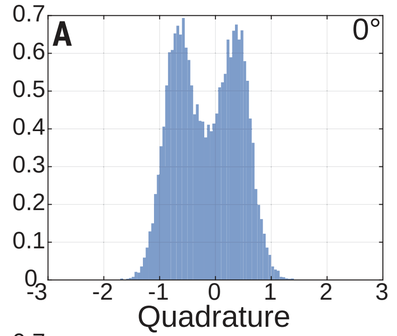
$|0_L\rangle$

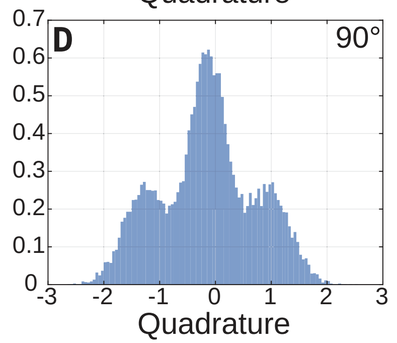
$|1_L\rangle$



Gottesman, Kitaev, Preskill, *PRA* **64**, 012310 (2001)

Realistic GKP states

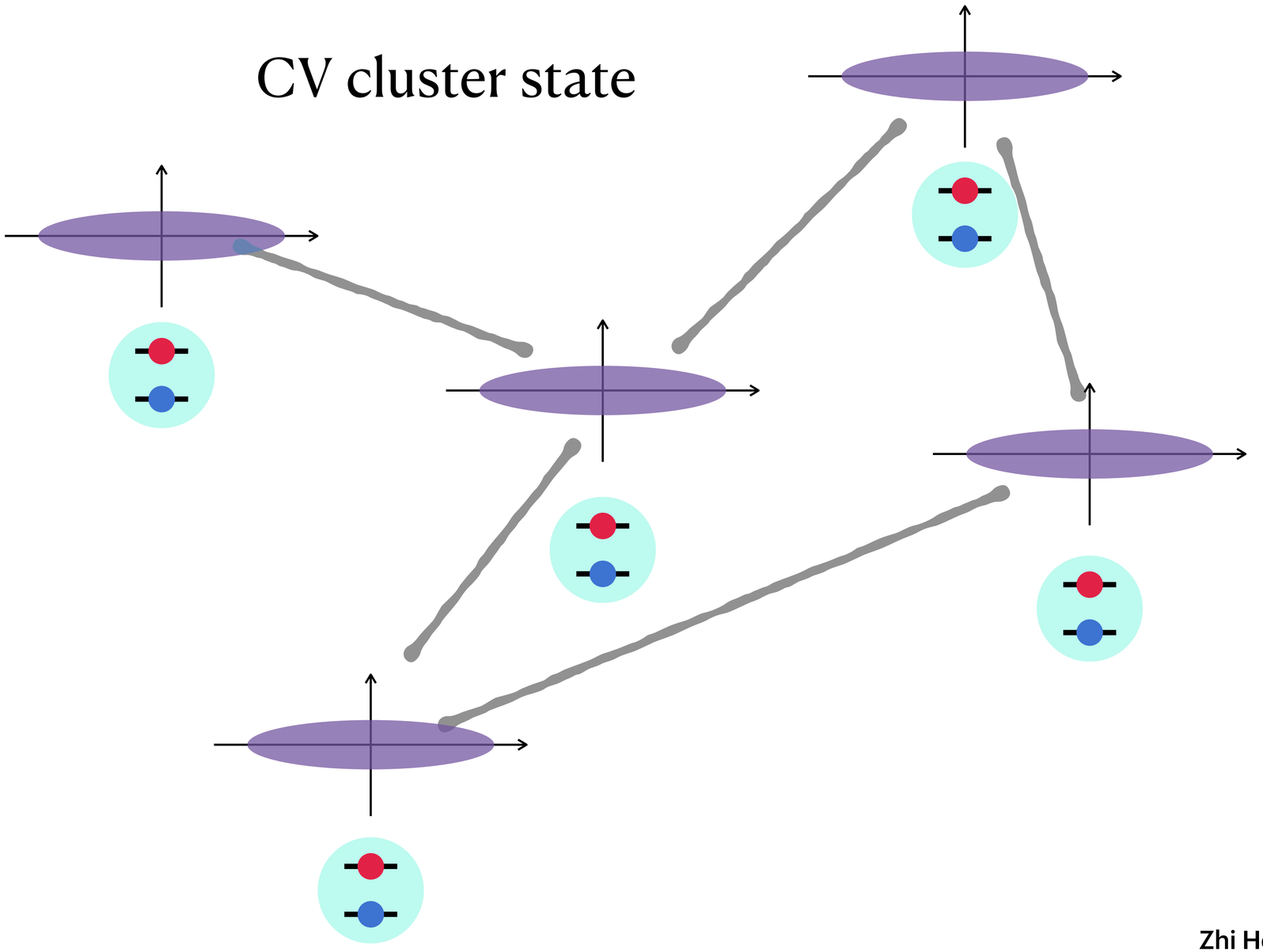




Konno *et al.*, *Science* **383**, 289 (2024)

- CV states
- Add noise

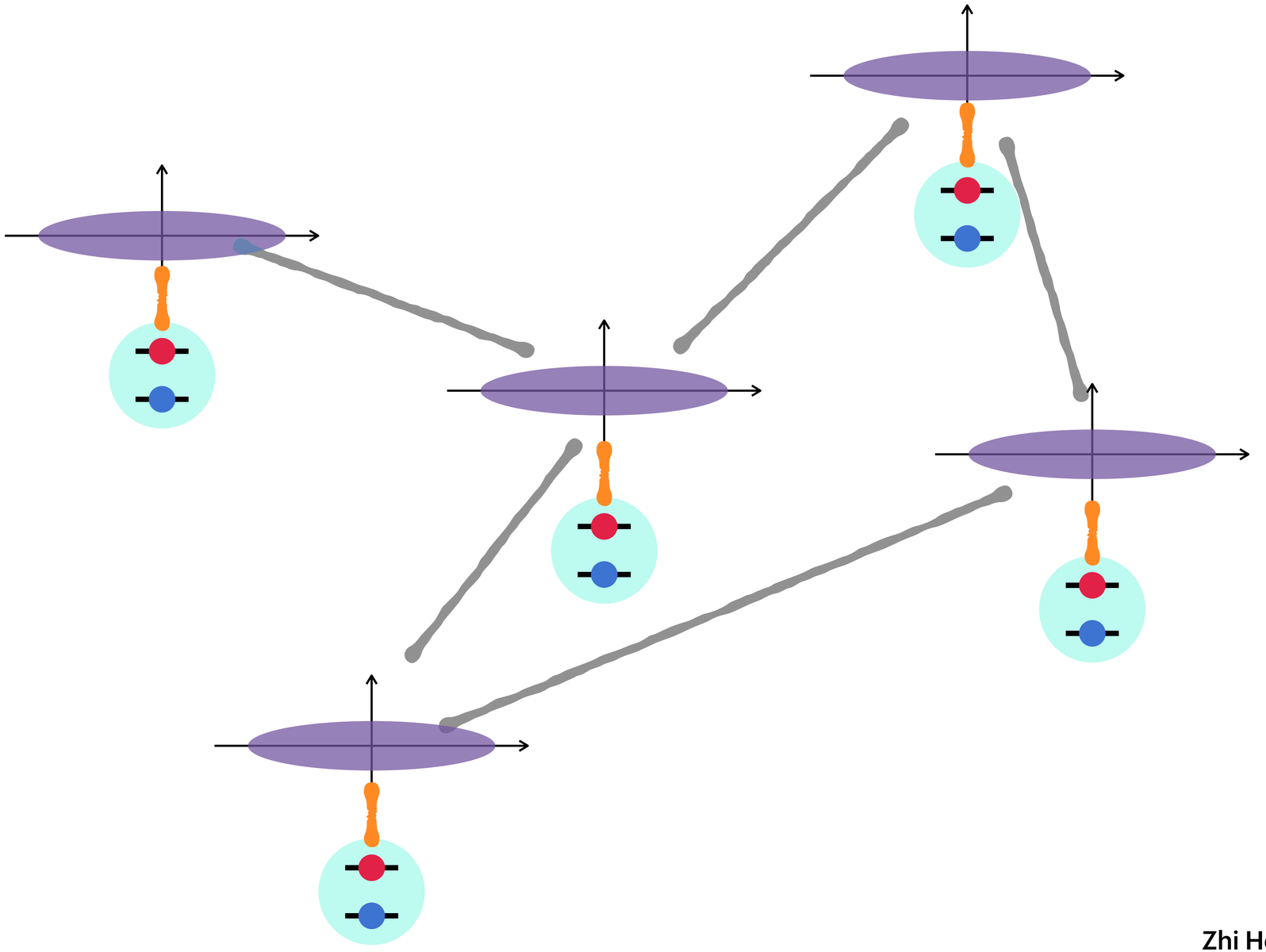
CV entanglement → physical qubits



1. Attach ancilla qubit to each mode

Zhi Han, Hoi-Kwan Lau, *Physical Review Letters* **136**, 040202 (2026)

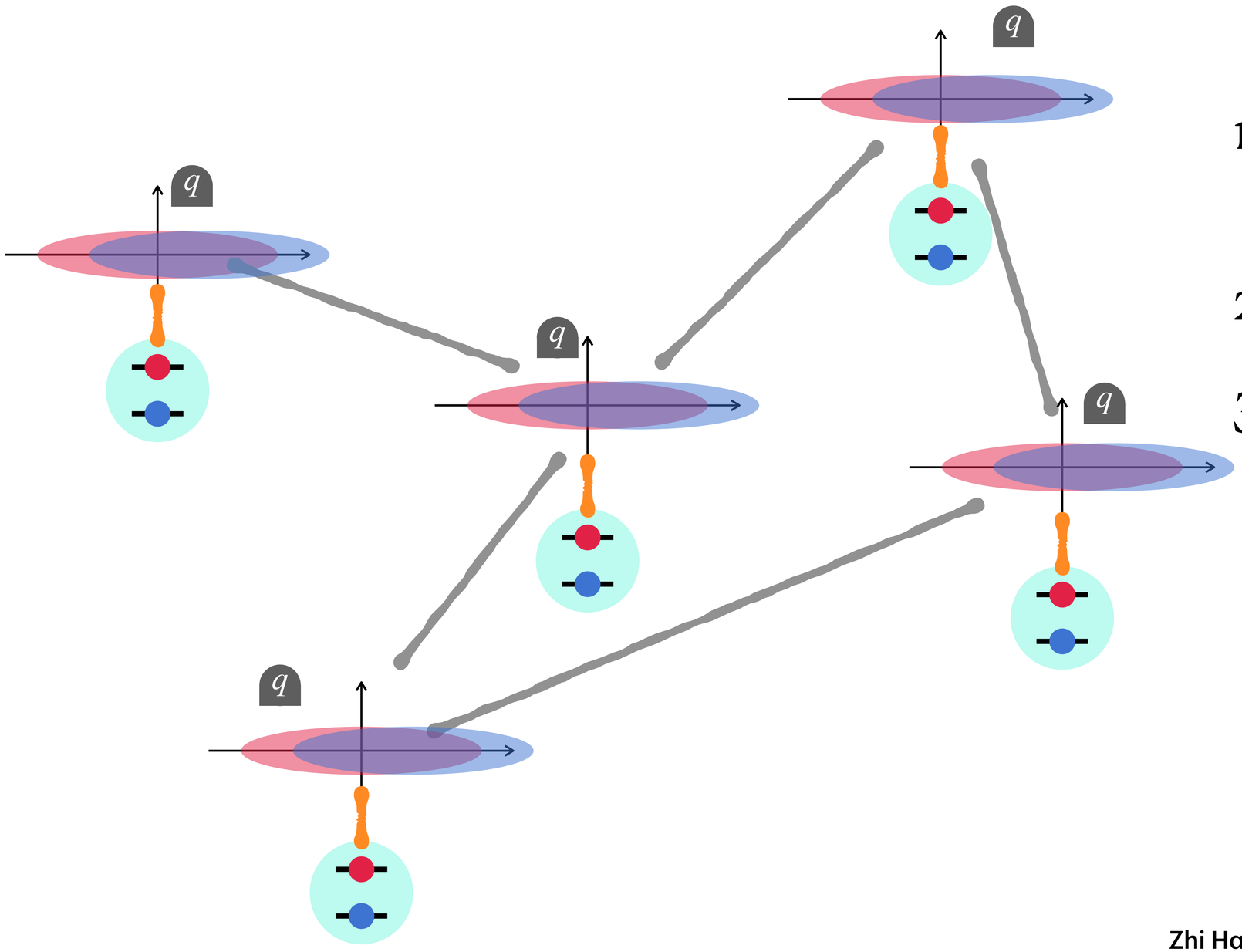
CV entanglement → physical qubits



1. Attach ancilla qubit to each mode
2. Conditional displacement

Zhi Han, Hoi-Kwan Lau, *Physical Review Letters* **136**, 040202 (2026)

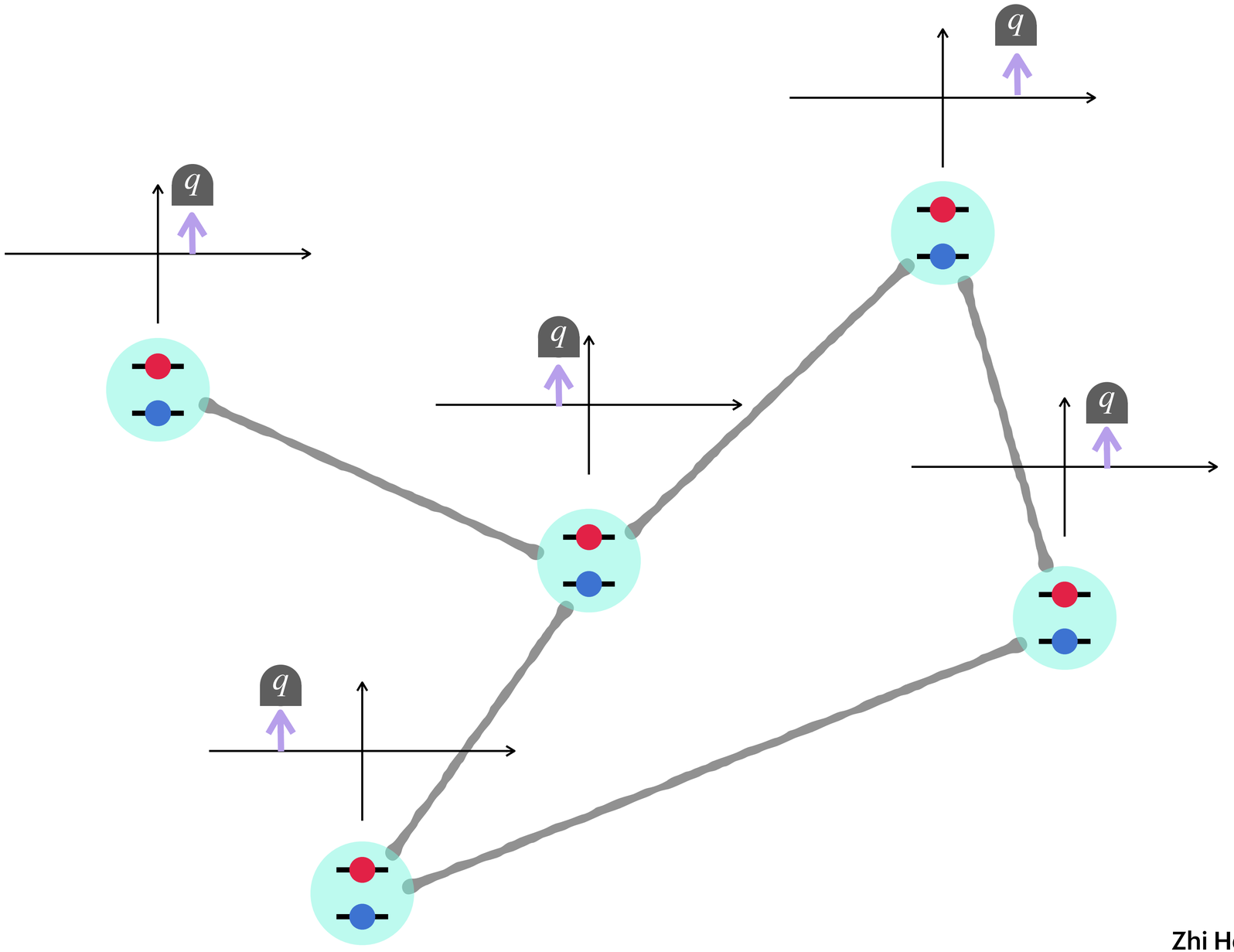
CV entanglement → physical qubits



1. Attach ancilla qubit to each mode
2. Conditional displacement
3. Homodyne detection

Zhi Han, Hoi-Kwan Lau, *Physical Review Letters* **136**, 040202 (2026)

CV entanglement → physical qubits



1. Attach ancilla qubit to each mode
2. Conditional displacement
3. Homodyne detection

Qubit cluster state created

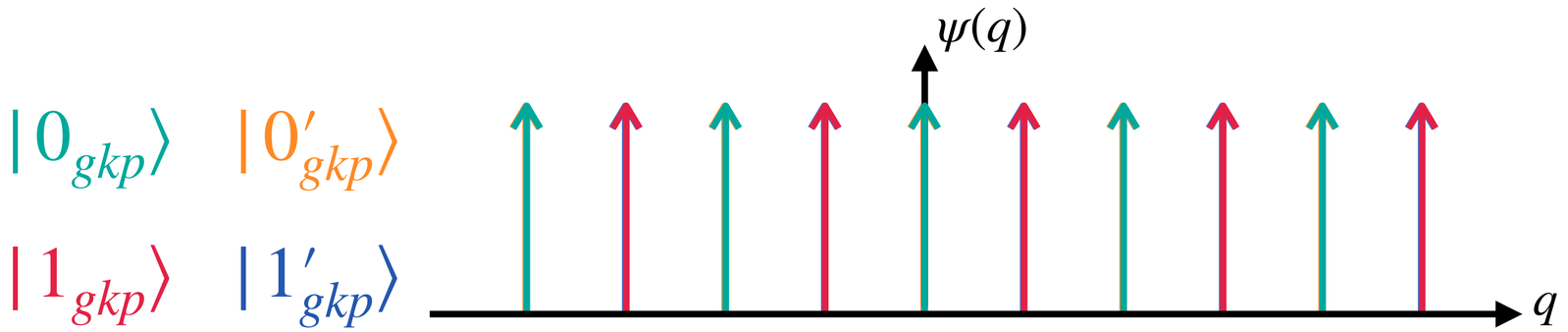


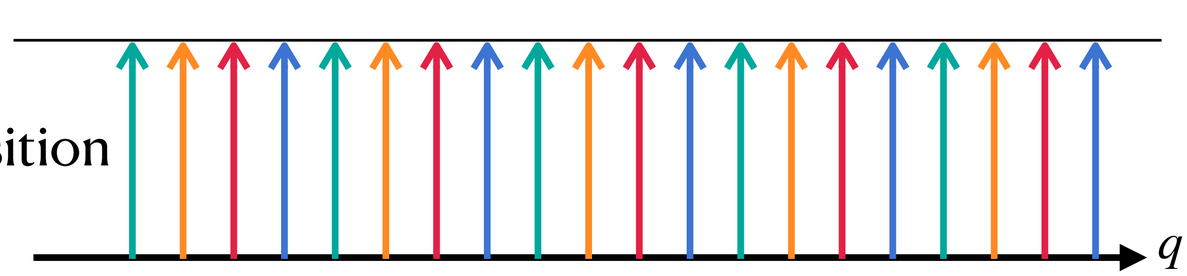
Zhi Han, Hoi-Kwan Lau, *Physical Review Letters* **136**, 040202 (2026)

Why there is qubit resource in CV cluster state?

Why does our scheme work (ideally)?

1. Infinitely squeezed state as superposition of displaced GKP $|+\rangle$





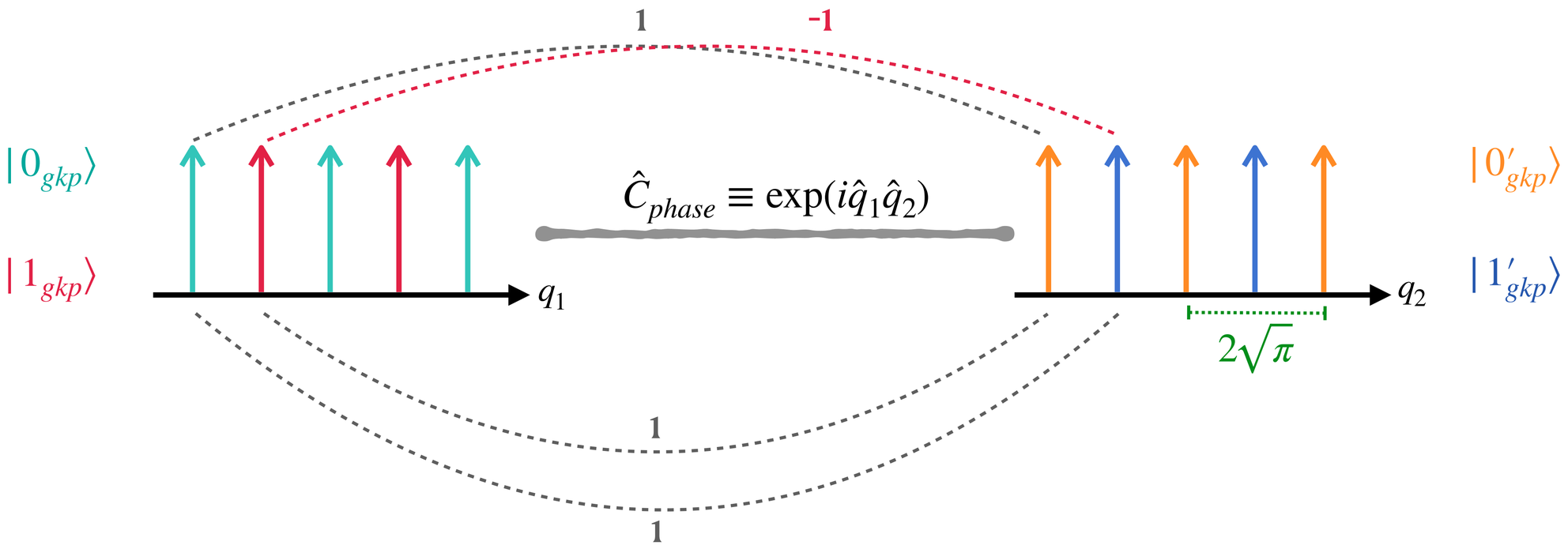
Infinitely p-squeezed state

$\psi(q) = \text{const.}$

Glancy & Knill, *PRA* **73**, 042304 (2006)

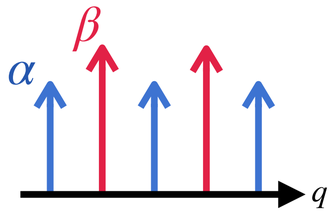
Why does our scheme work (ideally)?

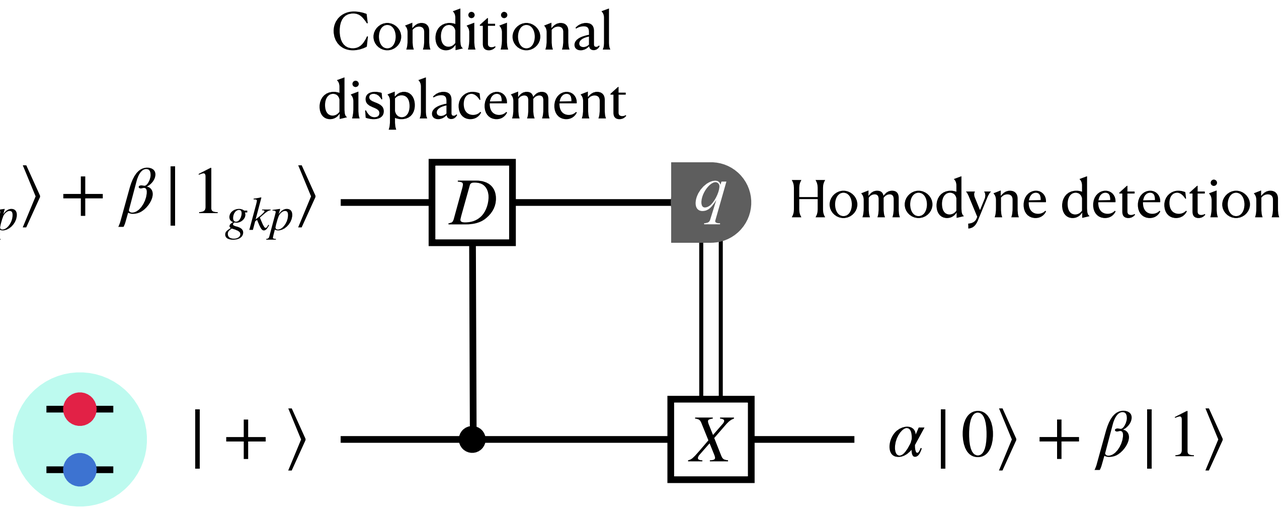
2. CV CPhase gate is CZ gate for qubits in displaced GKP basis



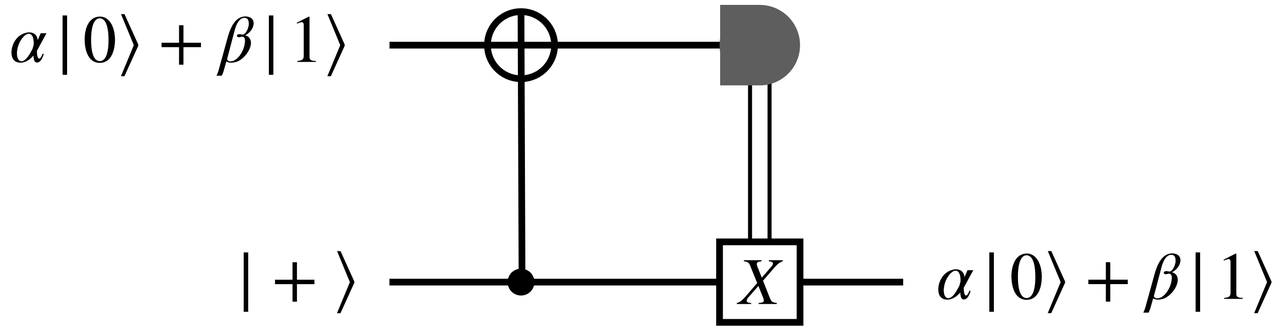
Why does our scheme work (ideally)?

3. Teleporting QI in GKP states to physical qubits

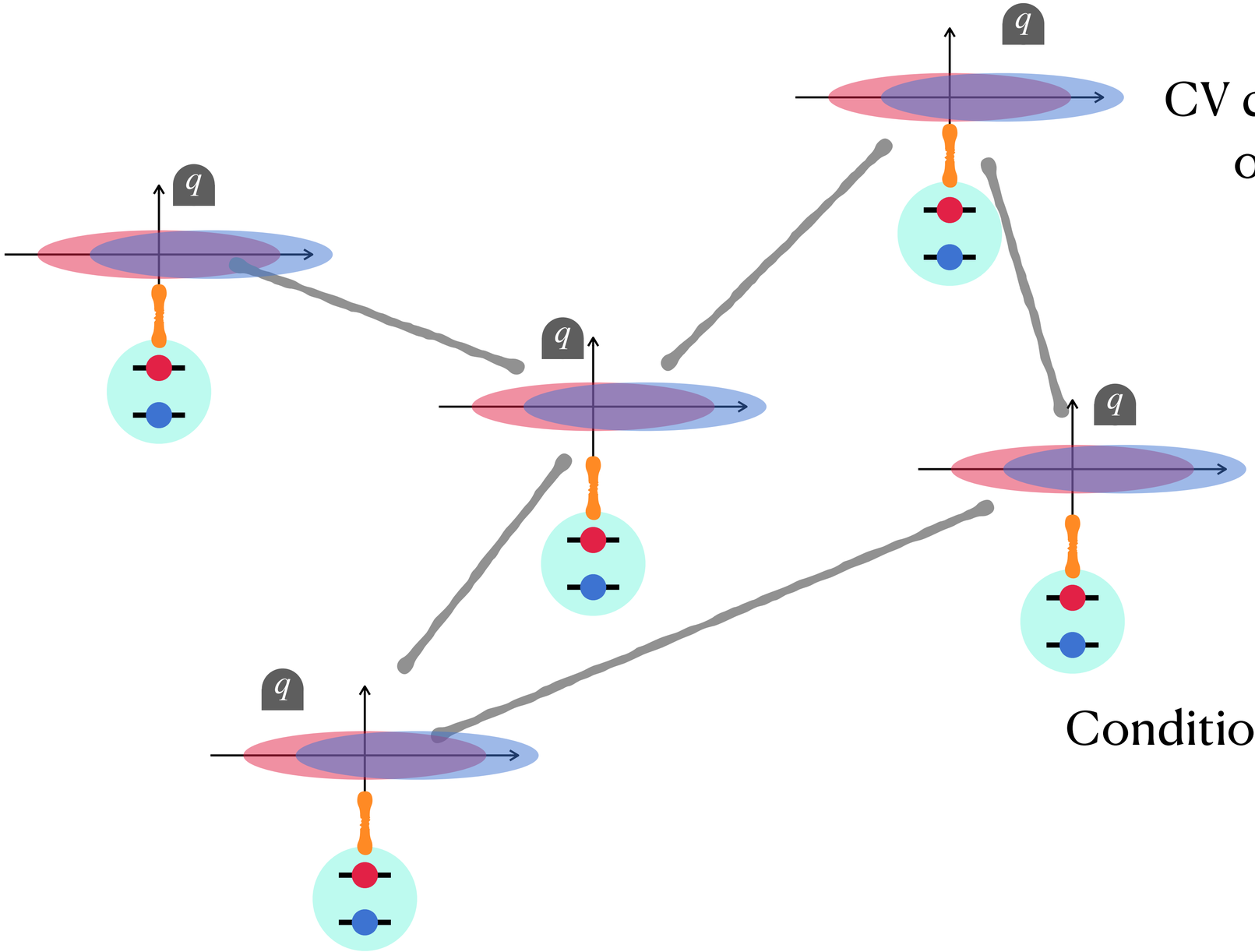




1-qubit teleportation



Why does our scheme work (ideally)?

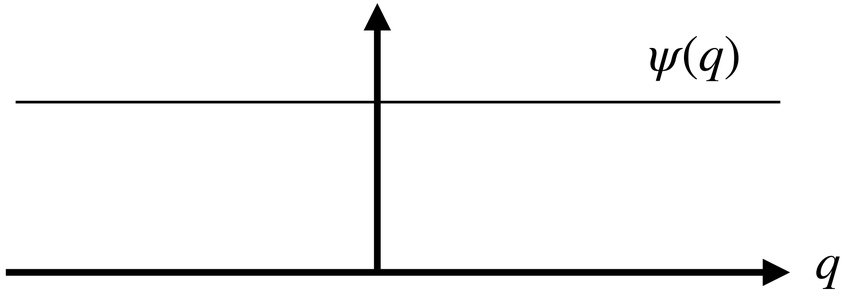


CV cluster state = Superposition
of GKP qubit cluster state

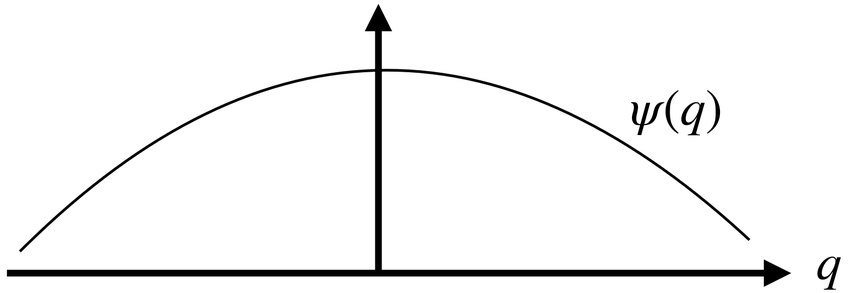
Conditional displacement + homodyne
= teleportation

How does our scheme work in practice?

Challenge 1: Finite squeezing

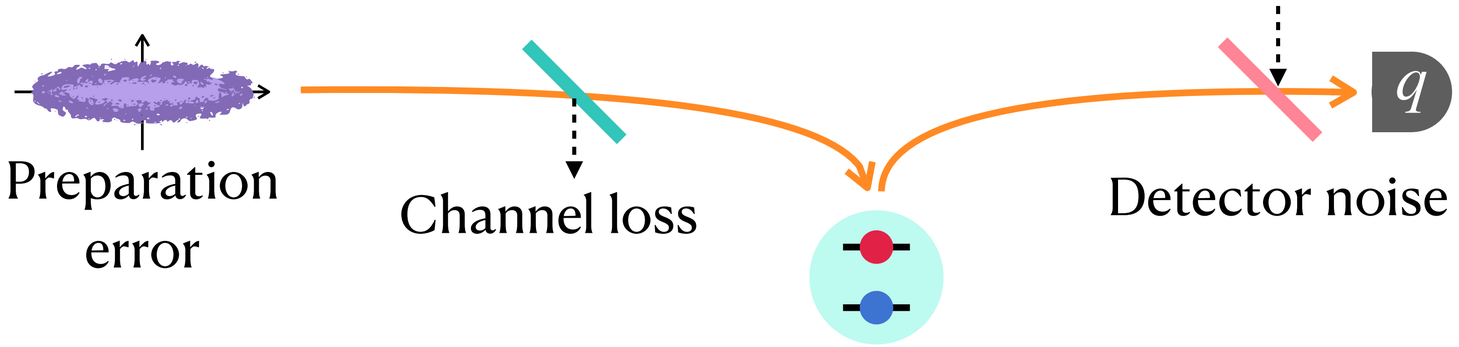


Ideal: infinite squeezing, infinite energy

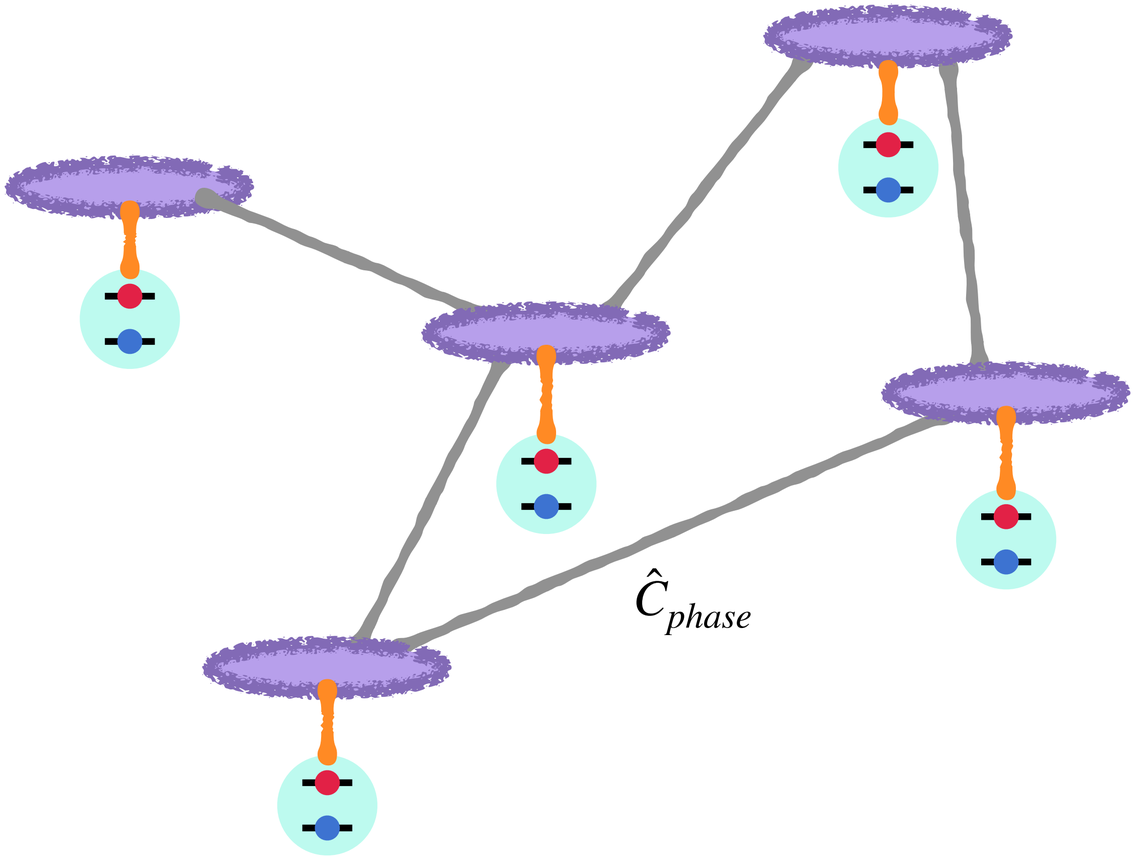


Practice: finite squeezing

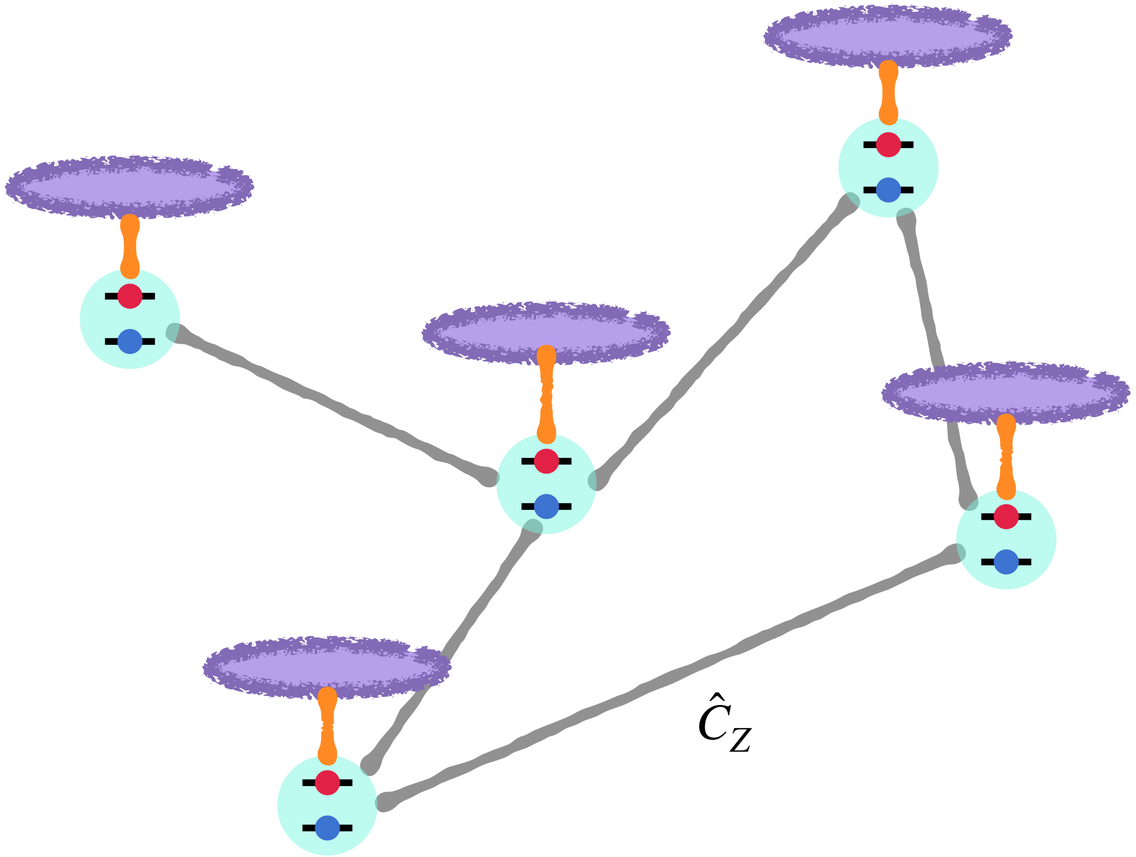
Challenge 2: Imperfection



Equivalent circuit model



=

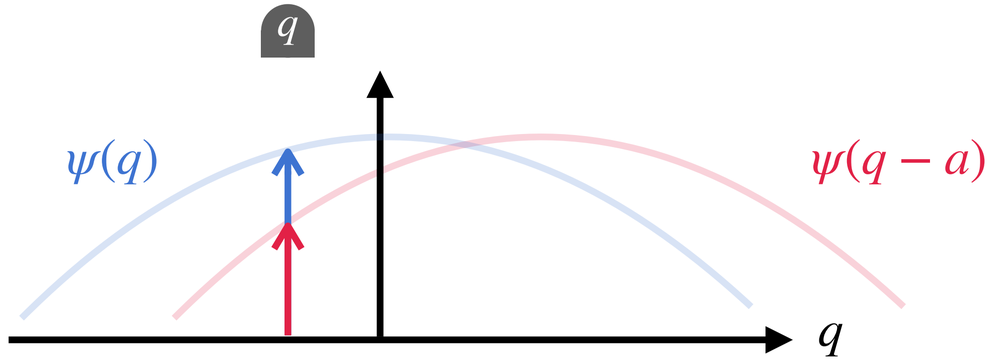


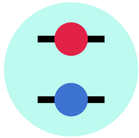
Qubits interact with imperfect finitely squeezed CV cluster state

Qubits corrupted by squeezed thermal state before perfect qubit gates

Single-qubit preparation error

Finite squeezing



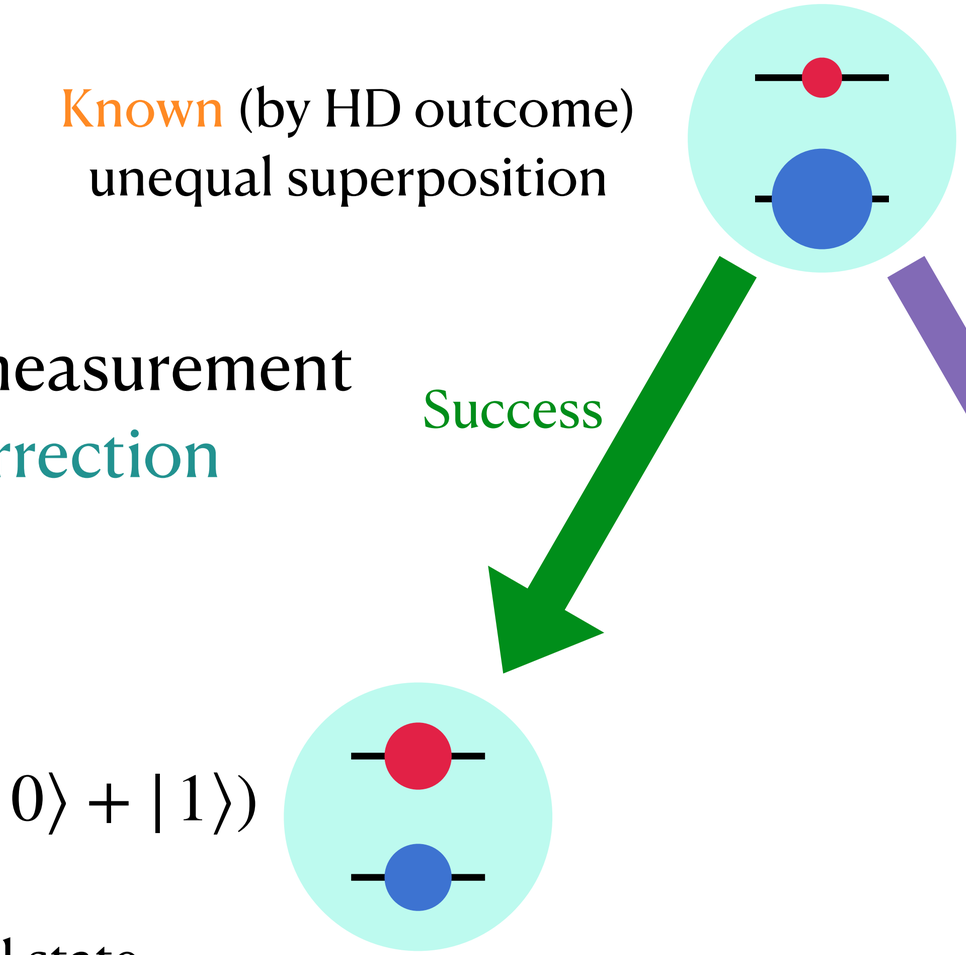


$$|\psi\rangle \propto \psi(q)\,|0\rangle + \psi(q-a)\,|1\rangle$$

Finite squeezing error

= Each qubit prepared in unequal superposition instead of $|+\rangle$

Probabilistic Correction



$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

Ideal initial state

$|0\rangle$

Computational basis

Finite squeezing error → probabilistic site erasure

Squeezing, as a resource, how much do we need?

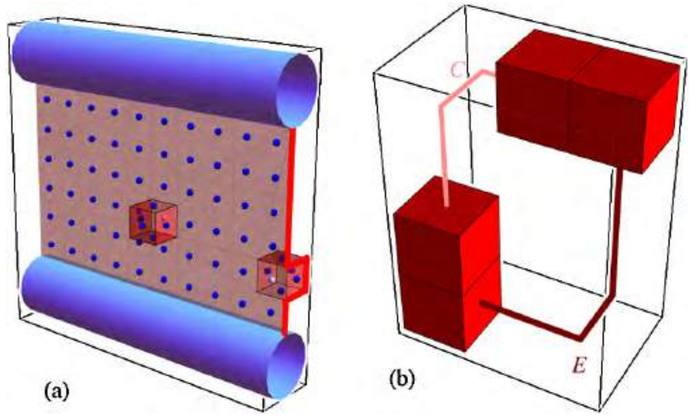
How much squeezing do we need?

Barrett & Stace, *PRL* **105**, 200502 (2010)

Fault tolerant quantum computing:

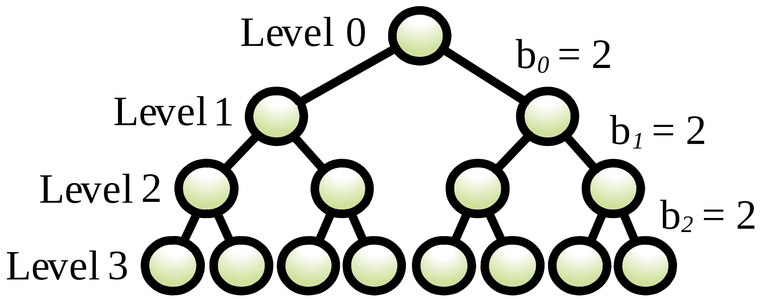
24.9% qubit loss ↔ 12 dB squeezing

Comparable to known squeezing threshold (10-12 dB) of GKP states

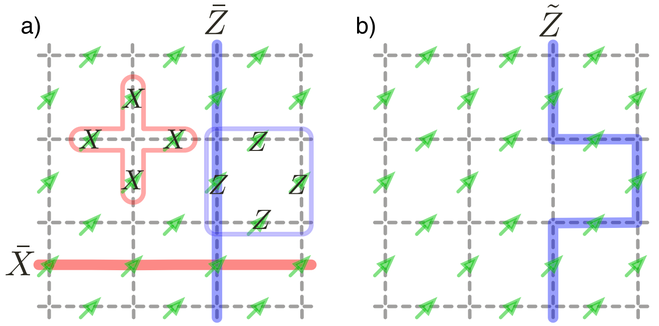


Quantum computing or robust memory: 50% qubit loss ↔ 6 dB squeezing

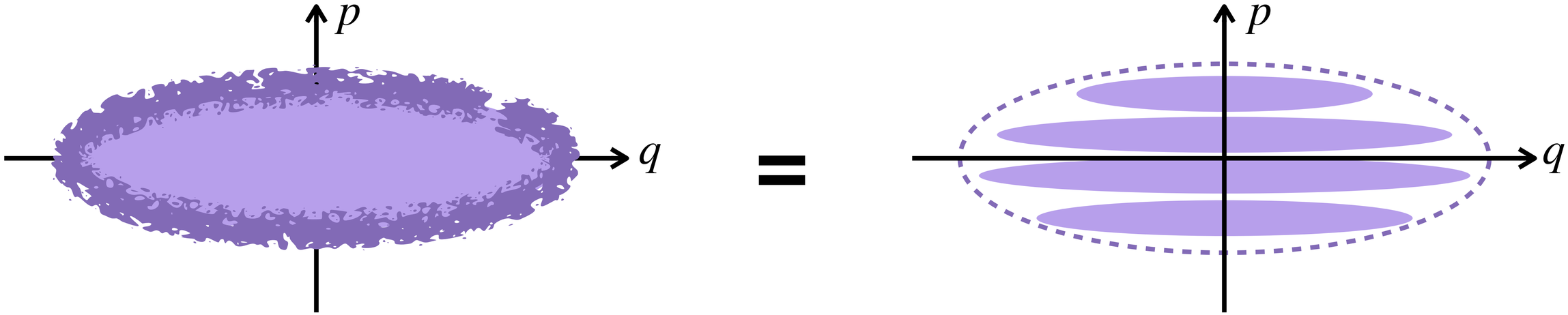
Varnava, Browne, Rudolph, *PRL* **97**, 120501 (2006)



Stace, Barrett, Doherty, *PRL* **102**, 200501 (2009)



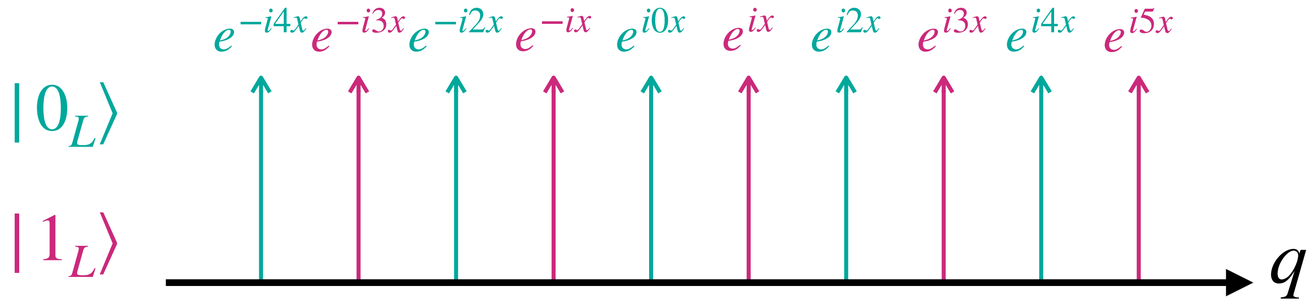
Physical imperfection



Squeezed thermal state

Randomly displaced pure squeezed state

Displacement operator: $\exp(ix\hat{q})$

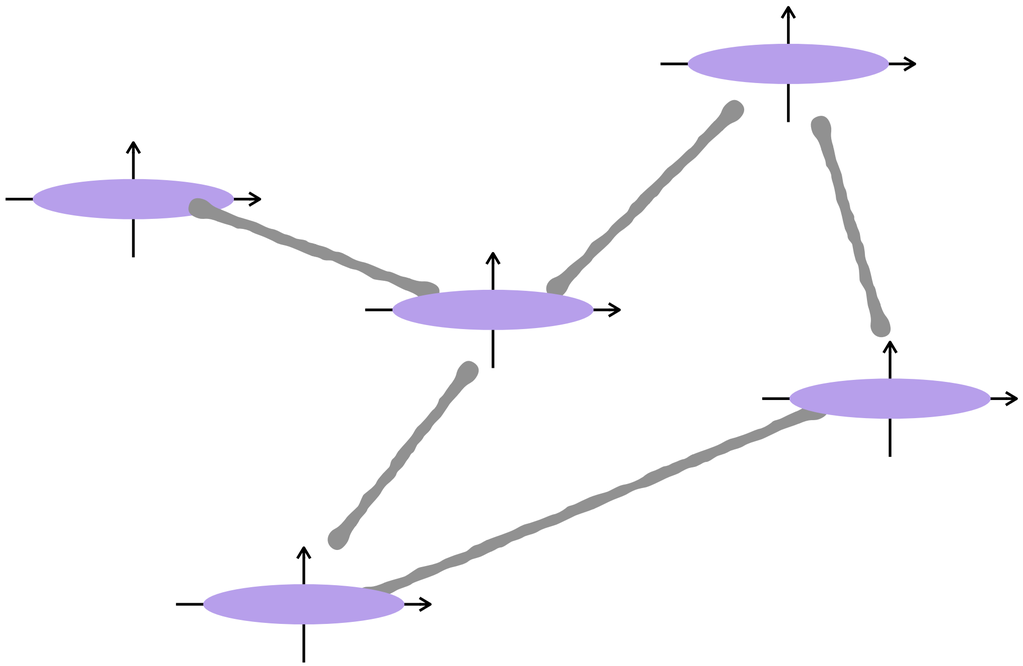


Preparation error + channel loss + detector inefficiency = single-qubit dephasing

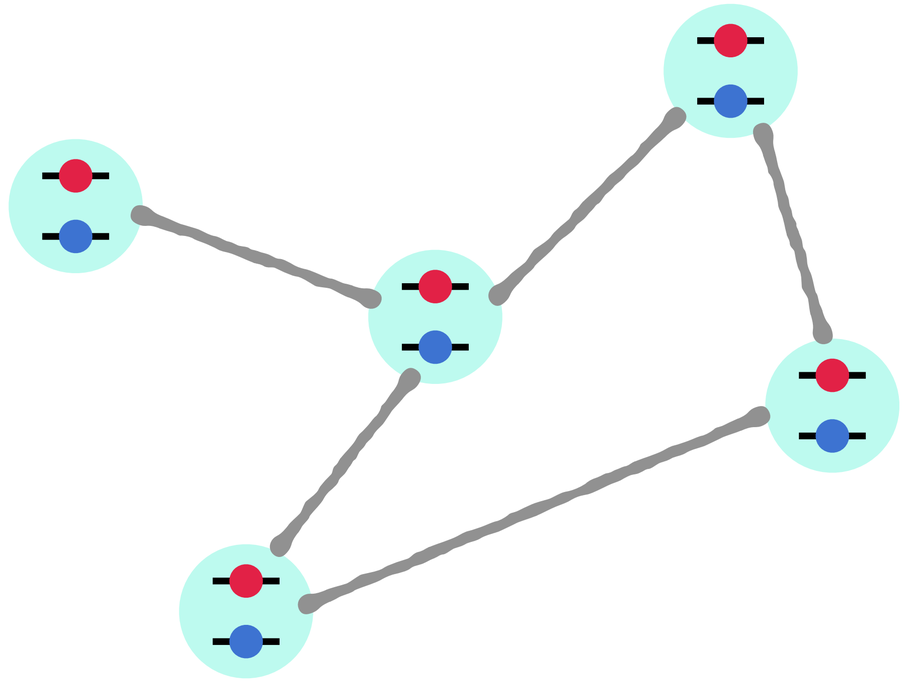
SFU

Downloading many-qubit entanglement from continuous-variable cluster state

CV cluster state (efficiently generated)



Qubit cluster state (lots of applications)



Equivalent circuit model: finite squeezing + imperfection → single qubit error

Realistic interaction: many possible implementation platforms

Zhi Han, Hoi-Kwan Lau, *Physical Review Letters* **136**, 040202 (2026)

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